Exploring the relationship between surfactants and swarming in uropathogenic *E. coli*

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Introduction

Escherichia coli strain CFT073, a cause of urinary tract infections (UTIs), uses flagella for movement. This includes individual swimming in liquid and coordinated swarming across surfaces (Fig 1). Swarming is important for virulence, especially in catheter-associated UTIs. *E. coli* CFT073 mutants lacking the gene *tolC*, which encodes an outer membrane efflux pump that removes compounds from the cell, cannot swarm, although they still swim. We hypothesize that the TolC system exports a factor required for surface movement, potentially a surfactant to reduce surface tension. To test whether a surfactant would rescue the non-swarming phenotype of Δ*tolC*, we tested swarming of WT and tolC mutants in plates with increasing concentrations of the detergent Triton X-100.

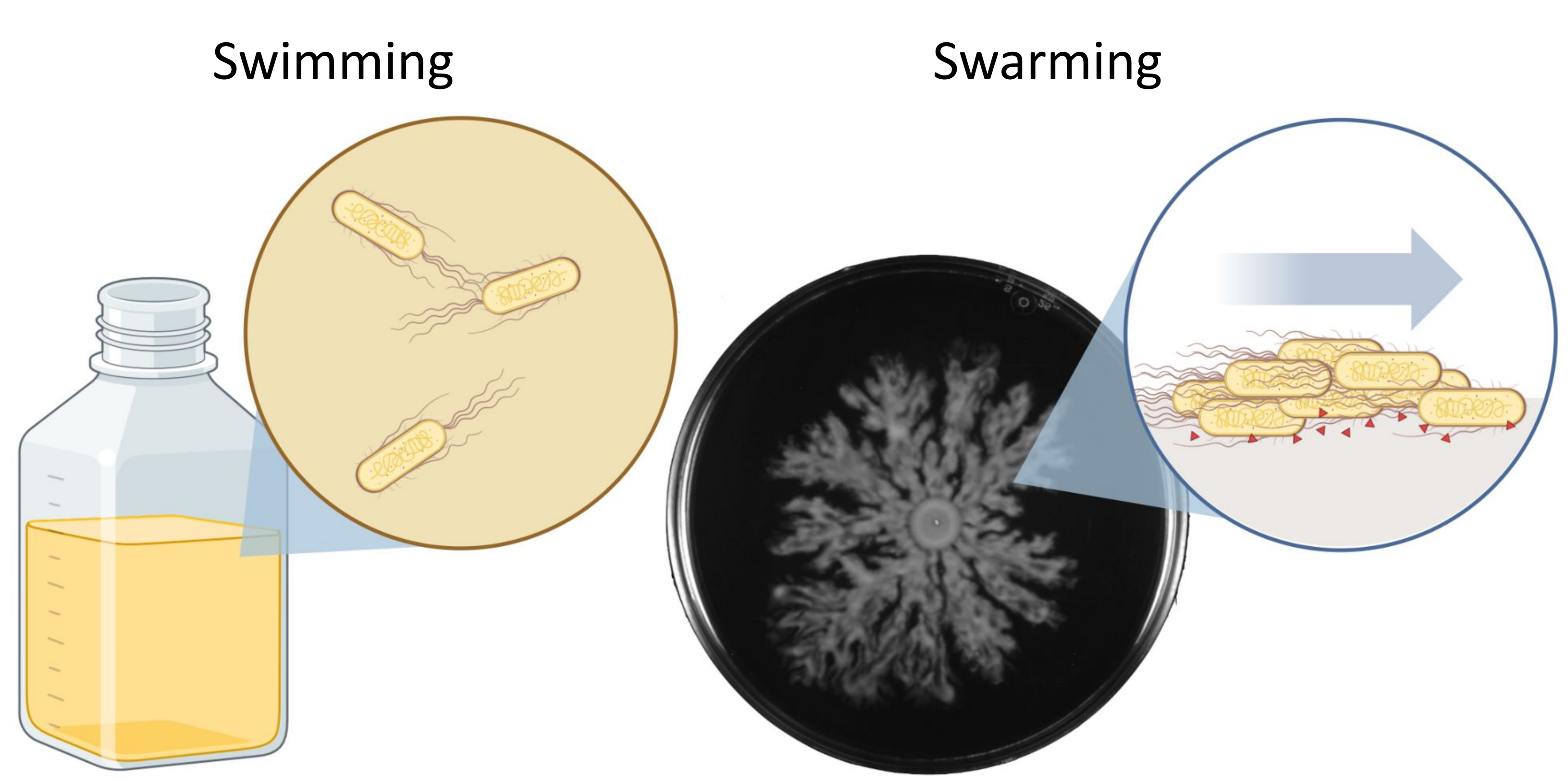

Figure 1. While individual WT *E. coli* expresses swimming motility in liquid media (shown on the left), multicellular swarming motility is expressed in solid media (shown on the right). Swarming often occurs in coordinated cell groups known as rafts, with many bacterial species secreting surfactants (red triangles) that lower the surface tension between the substrate and the cell allowing for successful spreading. This image was created with Biorender.com.

Materials and Methods

1. Make swarming media with 0%, 0.0001%, 0.0003%, 0.001%, 0.003% Triton X-100
2. Pour and Dry Plates Overnight
3. *E. coli* CFT073 WT and Δ*tolC* was grown overnight in Tryptic Soy Broth (TSB) at 37°C
4. Culture was taken to an OD600 = 1
5. Culture was spot inoculated on the plates and incubated for 24 hours
6. Plates were imaged on day 3. Swarm area was quantified using ImageJ software

Results

Without Triton X-100, WT expressed the expected swarming phenotype while Δ*tolC* expressed a non-swarming phenotype (Fig 2).

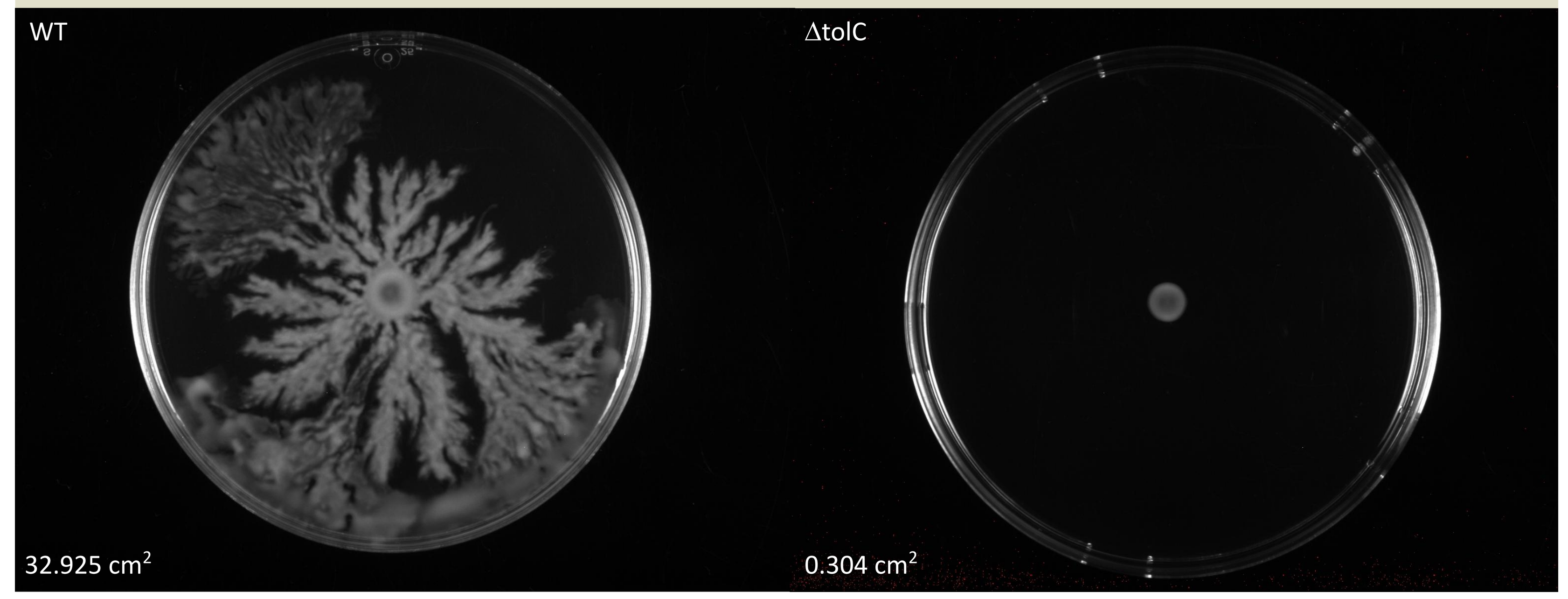

Figure 2. WT and Δ*tolC* grow on 0% Triton X-100, with WT expressing the standard swarming phenotype as expected and Δ*tolC* expressing the non-swarming phenotype.

At 0.0001% Triton X-100, the swarming behaviour of WT remains unchanged, with a slight increase in growth and some indication of swarming in its nascent stage occurring on Δ*tolC* (Fig 3).

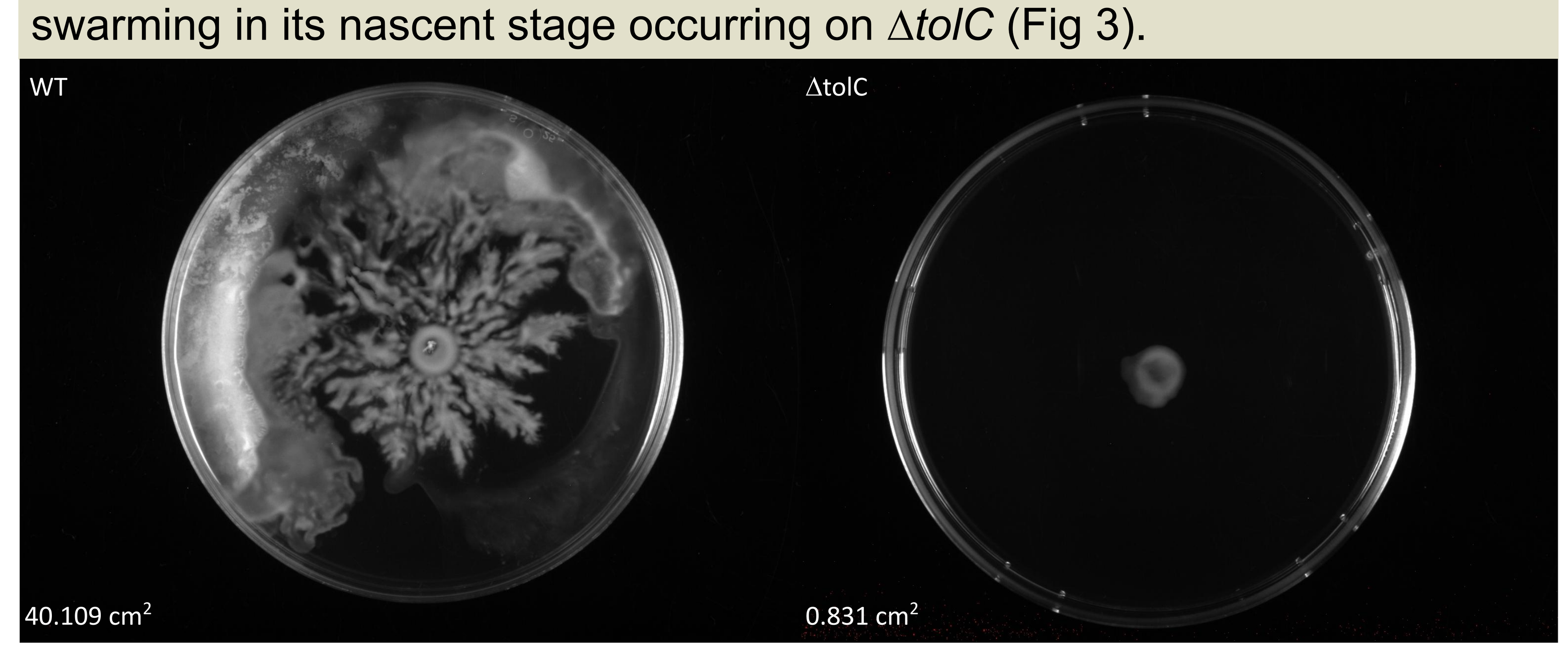

Figure 3. WT and Δ*tolC* grow on 0% Triton X-100, with WT swarming as expected and Δ*tolC* expressing the beginnings of what seems like swarming motility.

At 0.003% Triton X-100, WT expresses a similar swarming behavior as with lower surfactant concentrations. The non-swarming Δ*tolC* is able to initiate swarming, showing that a swarming phenotype can be restored by addition of external surfactants (Fig 4).

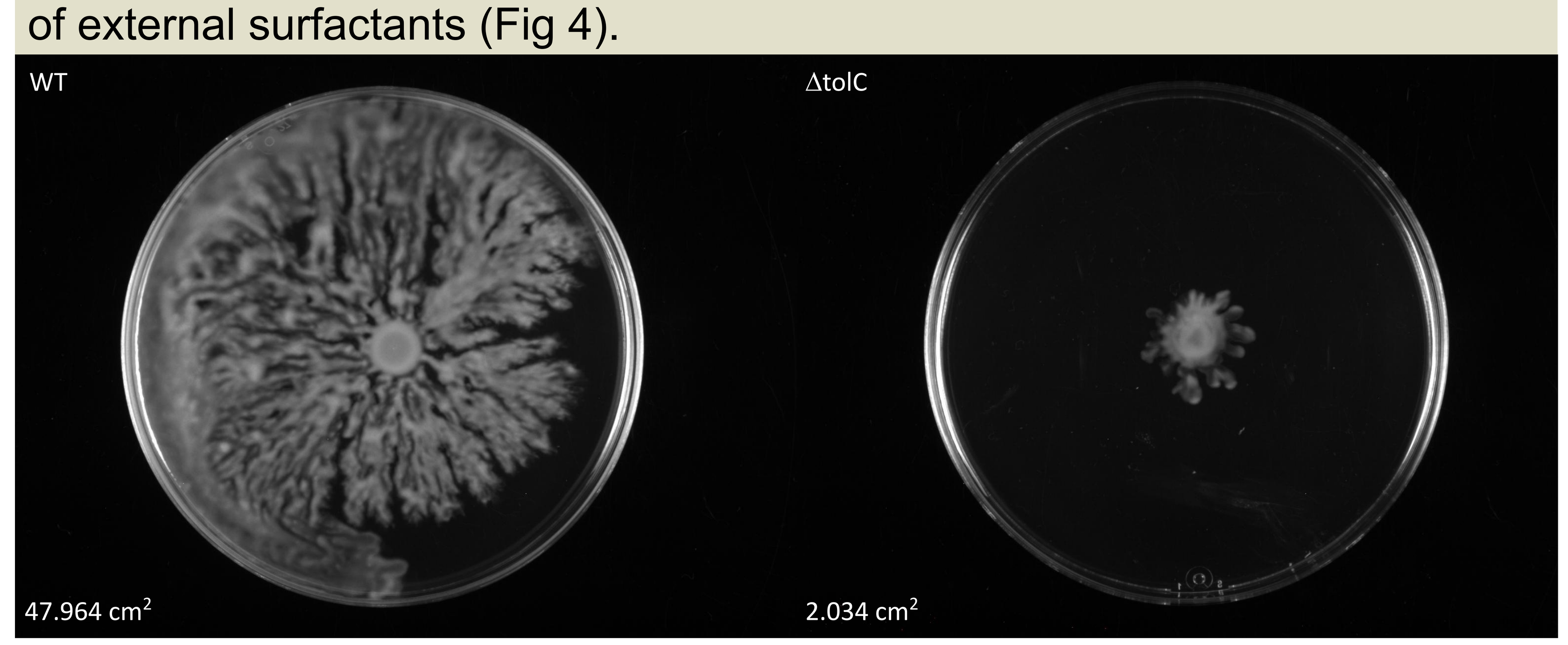

Figure 4. WT and Δ*tolC* grow on 0% Triton X-100, with WT swarming as expected and Δ*tolC* being rescued of its non-swarming phenotype by addition of Triton X-100.

Discussion

- We initially hypothesized that the introduction of a surfactant would rescue the non-swarming phenotype observed in Δ*tolC*
- When the concentration of Triton X-100 increased from 0% to, ultimately, 0.003%, the area of swarming on the Δ*tolC* plates increased
- Motility expressed on the WT plates largely remained the same.
- From these results, we can infer that the lack of a surfactant is playing a role in the non-swarming phenotype observed in Δ*tolC*
- Deletion mutant Δ*macA*, encoding a multidrug transporter that partners with TolC, also fail to swarm but retain the ability to swim. This suggests the TolC-MacAB efflux system may secrete a surfactant needed for swarming (Fig 6), (Inoue et al., 2007)
- Colanic acid was recently identified as a molecule conferring surfactant-like properties essential for swarming (Hwang et al., 2025)
- A deletion mutant Δ*wcaF* without one of the enzymes responsible for the biosynthesis of colanic acid was found through literature review to cause a non-swarming phenotype (Fig 6), (Hwang et al., 2025)
- We plan on formulating experiments involving colanic acid and Δ*macA*

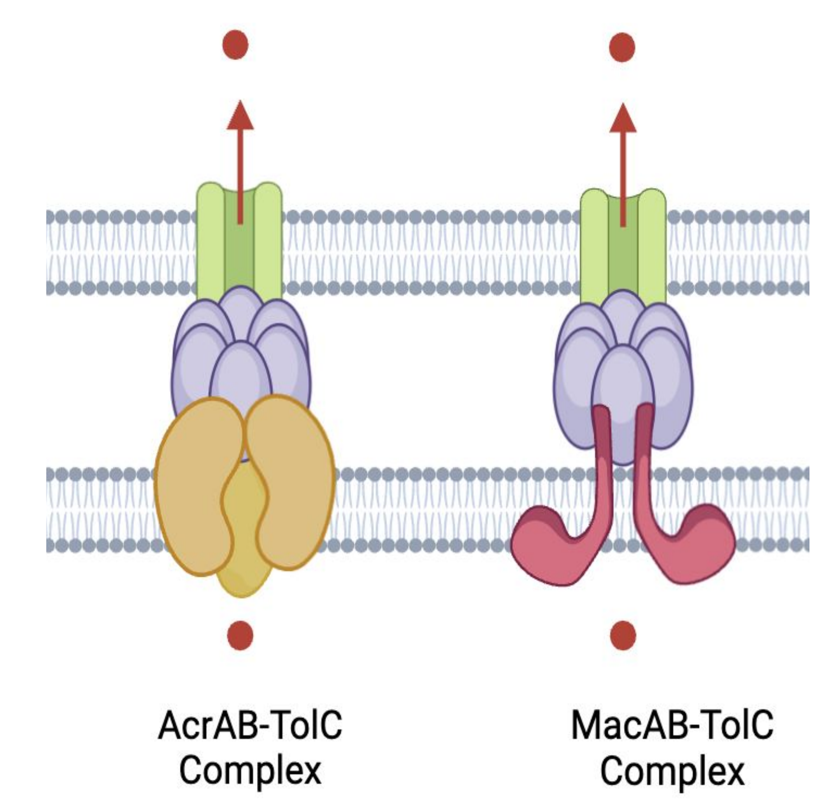

Figure 5. The AcrAB-TolC efflux pump is embedded in the cell membrane of gram negative bacteria, facilitating in the export of molecules out of the cell. Similarly positioned, the MacAB-TolC efflux pump is known as a multidrug transporter. The TolC protein is located in the outer membrane of *E. coli*. This image was created with Biorender.com.

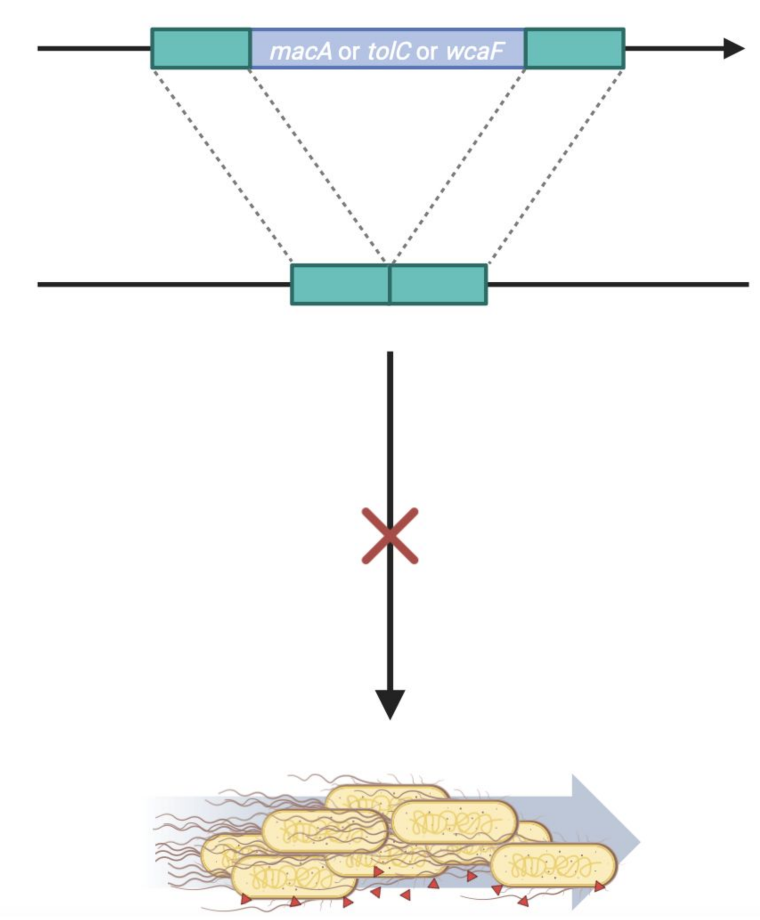

Figure 6. The deletion mutants Δ*macA*, Δ*tolC*, and Δ*wcaF* express the non-swarming phenotype.

References & Acknowledgements

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Inoue, T., Shingaki, R., Hirose, S., Waki, K., Mori, H., & Fukui, K. (2007). Genome-wide screening of genes required for swarming motility in Escherichia coli K-12. Journal of Bacteriology, 189(3), 950–957.

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